Heart Disease Risk Modelling via Bayesian Networks and Feed Forward Neural Networks Phase 2: Evaluation and Exploratory Hybrid Improvement

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Abstract—We compare a discrete domain-structured Bayesian network (BN), Naive Bayes (NB), a regularized feed-forward multilayer perceptron (MLP), and a BN-assisted MLP on historical UCI heart-disease records. The hybrid appends a cross-fitted BN posterior log-odds feature to continuous and one-hot predictors. Fixed configurations are evaluated on common stratified folds, with training-only epoch selection and three neural seeds. The standalone MLP is selected by development cross-validation. Its primary retrospective ROC-AUC is 0.9071, F1 is 0.8411, and Brier score is 0.1236. The hybrid obtains 0.9020, 0.8358, and 0.1252 respectively. In that original experiment, ablations and hospital holdouts do not establish a consistent hybrid advantage. A separate exploratory S-only nested-CV follow-up compares residual correction and calibrated fusion. The residual hybrid did not improve probability scoring. Its calibrated BN–MLP stack achieves Brier 0.13913 versus 0.14168 for tuned MLP and 0.14038 for calibrated BN, with improvement in every seed mean. The reported conditional interval includes zero and does not establish statistical significance. This is an exploratory observed Brier improvement, not proven superiority. The contribution is an implemented, leakage-aware comparison with auditable artifacts, rather than a claim of clinical utility or guaranteed hybrid improvement.

Index Terms—Bayesian network, MLP, cross-fitting, calibration, tabular classification

I. Introduction

The project investigates whether explicit probabilistic independence assumptions or learned continuous interactions better describe recorded disease presence. The term "risk" in the assigned title does not imply future-event prediction. The binary target is $Y = \mathbb{1}(\text{num} > 0)$, where the source values are 0–4. No model is clinically validated, and outputs are not treatment recommendations.

The AML phase supplied a small provisional BN and conventional classifiers. Phase 2 adds the requested pgmpy implementation, an NB structural ablation, a PyTorch MLP, and a technically motivated hybrid. The hypothesis is that a class-conditional probabilistic summary may improve a flexible learner on limited data, while continuous inputs recover interactions lost through discretization.

II. Related Work

Pearl provides the foundational treatment of graphical probabilistic reasoning [1]. The UCI documentation describes the disease-presence target and the processed Cleveland subset [2]. Gorishniy et al. demonstrate the importance of strong, consistently evaluated tabular neural baselines and find no universally superior learner [3]. Their ResNet and Transformer architectures are not implemented here. Grinsztajn et al. document competitive tree learners and tabular inductive biases [4], motivating our retained LR and RF controls. Dropout reduces co-adaptation [5], but whether it helps here is an empirical ablation. Guo et al. distinguish discrimination from calibration [6]; their findings motivate probability diagnostics, not an assumption that temperature scaling or any architecture guarantees clinical calibration.

Medical hybrid research provides useful context without validating this project's specific method. Zhao et al. combine a Bayesian-network branch with graph-convolutional reasoning for medical image diagnosis [8]. Our tabular logistic stack has neither their image attributes nor attention-coupled graph network. Weng et al. evaluate neural and other learners for ten-year cardiovascular events using routine UK clinical data [9]; our task classifies recorded disease presence and cannot inherit their prospective claims. The Super Learner framework uses cross-validation to combine predictors [10]; our penalized two-logit stack is narrower and makes no claim to all its theory. Van Calster et al. emphasize calibration in medical prediction [11]; our Brier and sparse reliability diagnostics do not establish clinical calibration.

III. Data and Preprocessing

The supplied merged CSV contains 920 records from four hospitals; exact-record deduplication ignoring row ID retains 918 (508 positive, 410 negative). It is not the independently acquired 303-record processed Cleveland dataset. The secondary Cleveland analysis uses only that site's development records from the supplied CSV.

The numeric inputs are age, trestbps, chol, thalch and oldpeak. Categorical inputs are sex, cp, fbs, restecg, exang and slope. Source ID, hospital, ca, thal and the target are excluded. Hospital labels are used for evaluation grouping only. Zero pressure and cholesterol are encoded as missing. Numeric medians, indicators and scaling, plus categorical modes and one-hot vocabularies, are learned only from the permitted training subset. Unknown categories are ignored by the one-hot encoder. The BN uses its own fitted bins and marginalizes missing prediction-time evidence.

Row IDs are not verified patient identifiers. Exact duplicate removal does not prove patient independence. The merged CSV's full acquisition and linkage history is unresolved. The design keeps retained rows disjoint, which is weaker than a verified patient-disjoint split.

IV. Models and Theoretical Basis

A. Discrete BN and Naive Bayes

The existing provisional graph factorizes as

$$P(Y, \mathbf{X}) = P(Y)P(S \mid Y, O) \prod_{i \neq S} P(X_i \mid Y), \qquad (1)$$

where S is slope and O is oldpeak. The class-root directions specify a generative factorization; they are not medical causal assertions. The additional oldpeak–slope dependency and all omitted dependencies remain assumptions requiring domain review. NB removes this single additional edge under otherwise identical encoding and smoothing.

Training-only median imputation precedes numeric tertile discretization. Repeated cuts and extreme cuts are removed; cut boundaries use right-side search, with unbounded exterior intervals. Quantiles are statistical cut points, not clinical thresholds. Categorical modes use a fixed vocabulary and deterministic tie order. Every fitted BN exports cut points, fill values, graph and conditional tables.

pgmpy estimates Dirichlet-smoothed conditional tables:

$$\widehat{P}(X_i = k \mid \mathrm{pa}_i = j) = \frac{N_{ijk} + 1}{N_{ij} + K_i}. \qquad (2)$$

Our vectorized exact branch inference is checked independently against pgmpy's variable elimination, including missing evidence, and against the original AML BN. Training imputation is an approximation and is not missing-data EM.

B. Regularized MLP

The network is $d \to 32 \to 16 \to 1$, with ReLU hidden activations, dropout 0.2 after each hidden layer and a sigmoid output for inference. It optimizes the numerically stable BCE-with-logits objective

$$\mathcal{L} = -\frac{1}{n} \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]. \qquad (3)$$

AdamW uses learning rate 0.001, weight decay 0.001 and batch size 32. Maximum training is 150 epochs, with patience 20 and validation-BCE improvement tolerance 10^{-4}. The early-stop subset is 20% of permitted training rows. Its validation rows cannot fit preprocessing or BN state. After selecting the epoch count, all permitted training rows are used to refit a newly initialized model for that fixed number of epochs. No outer-fold validation, cutoff or historical evaluation records enter epoch selection.

A small MLP matches unordered tabular inputs without inventing temporal or spatial structure. Its parameter count remains substantial relative to the sample size; regularization mitigates but cannot eliminate overfitting.

C. Feature-Level Hybrid

For each training record, three-fold stratified cross-fitting produces $q_i = P_{\mathrm{BN}, -fold(i)}(Y = 1 \mid \mathbf{x}_i)$. The additional input is

$$z_i = \mathrm{standardize}\left(\log \frac{\widetilde{q}_i}{1 - \widetilde{q}_i} \right), \qquad (4)$$

where $\widetilde{q}_i$ is clipped to $[10^{-6}, 1 - 10^{-6}]$ only for this transform. The MLP learns $\sigma(f_\theta([\phi(\mathbf{x}_i), z_i]))$; this is feature stacking, not a probability average or a novel algorithm. Early-stop optimization inputs are cross-fitted within that subset alone. Inference uses a BN refitted on all permitted training rows. That sample-size difference creates a potential cross-fitting-to-inference distribution shift. The hybrid adds 32 first-layer weights. An NB-assisted MLP tests whether the proposed BN dependency adds value beyond a generic probabilistic summary.

V. Validation Protocol

The unchanged stratified seed-42 historical split yields development $D = 734$ and retrospective $T = 184$. Seed 43 splits D into selection $S = 550$ and cutoff $C = 184$. The historical evaluation partition was previously inspected in AML; it is not newly untouched.

Five common stratified S folds use seed 44. Neural seeds 42, 52 and 62 are all retained. Fixed primary families are BN, NB, MLP and BN-assisted MLP. Rank by mean fold F1 at cutoff 0.5, averaged over all declared seeds, followed by ROC-AUC and declared family order. Seed 42 is the predeclared primary saved model, not the best observed seed. This is fixed-configuration CV, whereas AML used nested parameter searches. Search budgets differ and the winning family score has selection optimism.

Final models fit S only. Each frozen model chooses a cutoff on C from 0.00–1.00 in steps of 0.01, maximizing F1; ties choose the value nearest 0.5, then the lower value. Predictions use $p \geq t$. Models and cutoffs are saved before T labels are scored, with no $S + C$ refit.

We report ROC-AUC, trapezoidal PR-AUC, average precision separately, sensitivity, specificity, F1, Brier score, log loss, confusion matrices and five-bin reliability with counts. Both default and cutoff-selected metrics are stored. Constant-prior probabilities and always-positive decisions are references. Paired 2,000 row-bootstrap differences hold fitted models and cutoffs fixed; they omit

TABLE I
Primary selection CV, fixed configurations

Model	F1	ROC-AUC
BN	0.8179	0.8780
Naive Bayes	0.8167	0.8740
MLP	0.8296	0.8686
BN + MLP	0.8283	0.8677

TABLE II
Retrospective results, seed 42, C-selected cutoffs

Model	F1	AUC	PR-AUC	Sens.	Spec.	Brier
BN	0.8098	0.8765	0.8936	0.8137	0.7561	0.1544
Naive Bayes	0.8103	0.8772	0.8958	0.9216	0.5610	0.1532
MLP	0.8411	0.9071	0.9192	0.8824	0.7317	0.1236
BN + MLP	0.8358	0.9020	0.9172	0.8235	0.8171	0.1252
Logistic regression	0.8520	0.9022	0.9138	0.9314	0.6829	0.1241
Random forest	0.8634	0.8950	0.9046	0.9608	0.6707	0.1289

training, family-selection and patient-cluster uncertainty and do not constitute formal superiority tests.

Four hospital holdouts use D alone. Each has independent local S/C training on the other hospitals, fixed settings and seed 42. Cleveland-only five-fold CV also excludes every T row. No pooled learned state enters these analyses.

VI. Results and Ablations

The MLP is selected under the declared criterion. Its seed-42 architecture uses 1377 parameters and 11 refit epochs; the hybrid uses 1409 parameters and 10 epochs. No retrospective ranking changes the family choice. The hybrid improves seed-42 specificity relative to the MLP (0.8171 versus 0.7317), while lowering sensitivity (0.8235 versus 0.8824). This operating-point tradeoff is not evidence of overall superiority. Hybrid-minus-MLP paired 95% intervals are [-0.0407, 0.0276] for F1, [-0.0157, 0.0058] for ROC-AUC, and [-0.0046, 0.0078] for Brier score. Each spans zero.

Removing dropout produces mean selection F1 0.8296, and removing weight decay produces 0.8310, versus 0.8296 for the base MLP. NB-assisted MLP reaches 0.8294 versus 0.8283 for BN-assisted MLP. These fixed-setting results do not show that every regularizer or the extra BN edge is beneficial. All seed-level results and both evaluation operating points are retained; none is chosen to improve the narrative.

Hospital transfer is uneven. MLP ROC-AUC is 0.8635 in Cleveland, 0.8791 in Hungary, 0.7372 in Switzerland and 0.5866 in VA Long Beach. The hybrid reaches 0.6930 in VA Long Beach, but specificity there remains 0.025 at its local cutoff, versus 0.050 for the MLP. A favorable AUC difference at one site does not demonstrate clinical utility or consistent hybrid value. Switzerland's high disease prevalence makes always-positive F1 a strong reference; saved site-level constant references must be consulted alongside model F1.

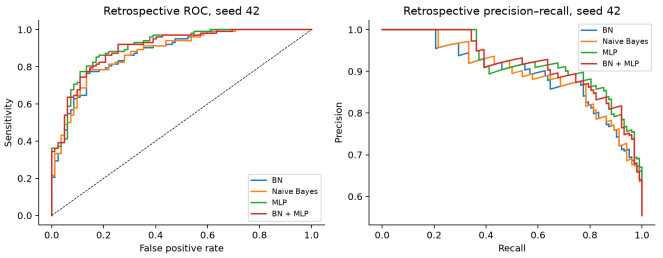


Fig. 1. Seed-42 retrospective discrimination. Curves summarize this historical partition only.

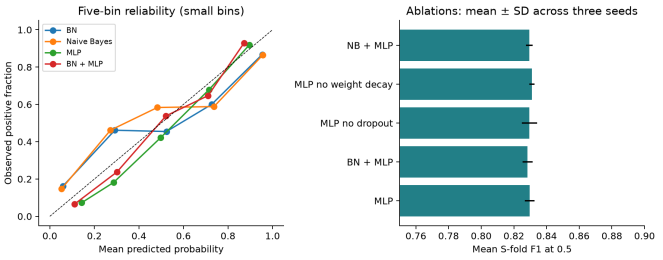


Fig. 2. Descriptive reliability and fixed-fold ablations. Error bars show standard deviation of seed-mean CV F1, not independent confidence intervals.

VII. Reproducibility and Technical Validation

The repository includes a dedicated Phase 2 dependency specification and run-specific version lock, source and protocol hashes, exact partitions, common folds, every seed's predictions, training curves, early-stop and cross-fitting audits, BN states and cut points, frozen models, model hashes, cutoff searches and reconstructable metrics. The source CSV and AML artifacts are preserved.

The full-run verifier checks 35 saved models, including saved-model probability round trips, cutoff reconstruction, metric reconstruction, family selection, CV coverage and hospital isolation. Local testing passed 60 tests with 39 subtests; the unchanged Phase 1 Streamlit interface was excluded from this Phase 2 environment check. GitHub automation runs the dedicated Phase 2 tests and a smoke experiment on pushes. Smoke runs use three epochs and must never be cited as performance evidence.

The application-independent batch inference command accepts the eleven eligible predictors, verifies model integrity and uses the saved cutoff without fitting. The original Phase 1 web interface retains its own model evidence. Only trusted joblib files should be loaded.

VIII. Original-Experiment Limitations

The experiment is small and retrospective, with uncertain patient linkage, incomplete merged-data provenance, hospital shifts and provisional graph assumptions. Row-level CV may exploit source-specific missingness. Calibration bins are sparse and descriptive. No prospective cohort, causal analysis, utility analysis or fairness validation is supplied.

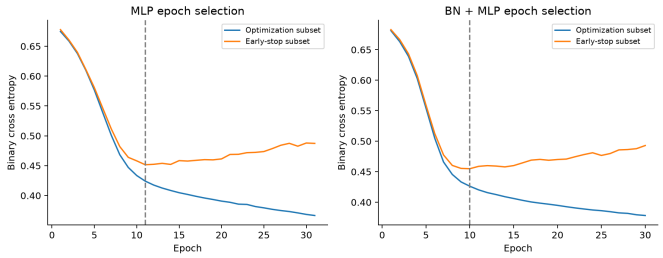


Fig. 3. Training-only epoch selection. Curves use the optimization and early-stop subsets within S, not T.

In the original experiment, the feature hybrid was implemented successfully but did not establish added value over the MLP. That original experiment alone does not establish performance-based hybrid value. The follow-up below tests improvements separately. The main deliverable is a transparent BN–MLP probability comparison with controlled preprocessing, ablations and reproducible evidence. Faculty approval of domain assumptions, correct individual contribution attribution, and the team's own presentation and viva remain essential.

IX. Exploratory Hybrid Improvement Study

A. Motivation and Prespecified Follow-up

The original BN-feature MLP did not improve the plain MLP consistently. After those results were inspected, we documented a finite follow-up protocol before its new results. This is exploratory research on reused development records, not confirmation on a newly untouched cohort. The original run, F1-first criterion and negative result are preserved. Only the 550 selection records S are used. The 184 cutoff records C and 184 retrospective records T are excluded from both fitting and scoring.

The follow-up evaluates two mechanistically different hybrids. The BN residual MLP starts at the BN log-odds and learns a shrunken continuous correction, with a zero-initialized output layer. The calibrated BN–MLP stack estimates

$$P(Y=1 \mid x) = \sigma\big(b + w_B \operatorname{logit}(q_B(x)) + w_M \operatorname{logit}(q_M(x))\big). \quad (5)$$

Unlike a fixed arithmetic mean of probabilities, this stack learns an intercept and penalized log-odds coefficients. The coefficients are fitted using fresh cross-fitted predictions of both base learners within the current training subset. They represent a statistical fusion of coarse structured evidence and learned continuous interactions, not causal feature effects or probabilities that sum to one. This is standard stacking, not an algorithmic novelty claim. It follows established out-of-fold stacking practice [7].

Calibrated BN alone and calibrated MLP alone are essential controls: without them, gains caused by calibration could be wrongly attributed to hybridization. A tuned MLP and the original feature hybrid are also retained.

TABLE III
Exploratory S-only nested outer results; all seeds

Model	Brier ↓	ROC-AUC	PR-AUC	F1
Calibrated BN-MLP	0.13913	0.87670	0.88437	0.82635
Calibrated BN	0.14038	0.87521	0.88280	0.82276
Original BN feature MLP	0.14150	0.87068	0.87741	0.82452
Tuned MLP	0.14168	0.86896	0.87384	0.82166
Calibrated MLP	0.14228	0.86883	0.87358	0.82174
BN residual MLP	0.14775	0.87191	0.87983	0.82247
BN	0.14868	0.87521	0.88280	0.82290

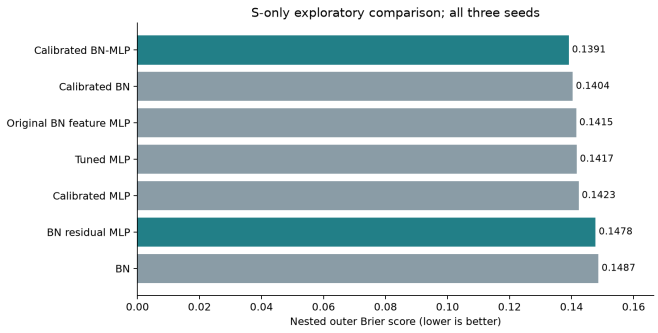


Fig. 4. Follow-up nested Brier comparison. Lower values indicate better mean probability scoring on S-only outer validation records.

Residual strength uses [0.25,0.5,1.0]; the equally budgeted MLP uses weight decay [0.0001,0.001,0.01]. Each logistic meta-model selects C from [0.1,1,10].

Five common outer stratified folds use seed 144. Three inner folds, seed 145, choose configurations by Brier score, then ROC-AUC and grid order. All neural seeds 42,52,62 are retained. Every inner training operation repeats preprocessing, early stopping and cross-fitting locally; no global out-of-fold matrix is reused across outer folds. The primary outcome is Brier because the follow-up explicitly tests probability quality. This is a declared change of research question, not a replacement for the original F1-first conclusion. F1, sensitivity and specificity use cutoff 0.5 only.

B. Follow-up Results and Ablations

The best evaluated hybrid is Calibrated BN-MLP, with mean outer Brier 0.13913. The strongest single-base control is Calibrated BN, with Brier 0.14038. The observed relative Brier reduction is 0.89%. The paired Brier difference is -0.00126; its conditional 95% row-bootstrap interval is [-0.00405, 0.00162]. The hybrid wins on Brier in 11 of 15 seed/fold pairs. Every seed improves on the strongest control.

The residual hybrid has mean Brier 0.14775, worse than tuned MLP (0.14168) and calibrated BN (0.14038); it did not improve probability scoring. Zero initialization starts the residual prediction at the BN baseline but does not guarantee improved generalization. The full tables retain all seven families and their unfavorable results. Both single-base calibrated controls and tuned MLP are

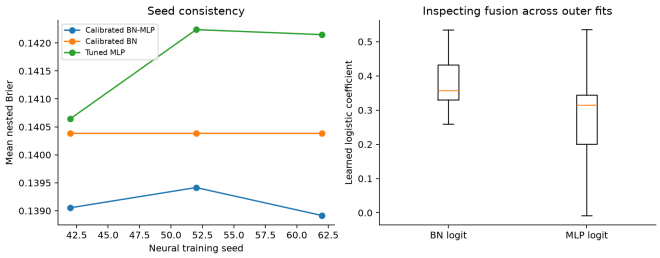


Fig. 5. Seed consistency and fitted stacking coefficients. Coefficients are log-odds weights, not causal effects.

compared with each hybrid. Removing either component tests the value of fusion, whereas the residual alternative tests a different architectural rationale. Claims must name the metric and comparator rather than imply uniform superiority on every measure.

C. Score Shift and Post-selection Uncertainty

OOF-to-full-fit score-distribution shift remains: the meta-model trains on out-of-fold base predictions, while inference uses base models refitted on the full permitted training subset. Their score distributions may differ. Cross-fitting prevents direct own-label leakage but cannot eliminate this sample-size shift.

Post-selection uncertainty also remains: multiple follow-up alternatives were evaluated, and the best was identified using the same outer-CV results later used to describe it. Inner tuning protects within-fold selection, but does not make that post-selected outer score independent confirmation. The conditional interval includes zero, so statistically significant superiority is not established under the reported uncertainty analysis. No formal post-selection significance test was performed.

D. Interpretation and Validation

The follow-up verifies 105 saved outer-fold models, including exact fold coverage, inner-grid selection, meta cross-fitting, early-stop isolation, source IDs, model hashes and probability round trips. Final inference models are refitted on S only after the study; their training round-trip predictions are not reported as performance estimates. No C/T scores are produced for these revised architectures.

Bootstrap resamples preserve source rows across all three seeds. The intervals condition on the saved outer predictions and omit overlapping-CV training dependence, patient clustering and research-selection uncertainty. The follow-up was initiated after earlier results; many comparisons remain exploratory. Observed improvements can support the technical value of the hybrid project, but do not establish external generalization, clinical utility, algorithmic novelty or guaranteed bonus marks. The graph, provenance and team-authorship qualifications still apply.

X. Overall Conclusion

The original feature hybrid and residual correction did not demonstrate improved probability scoring over their relevant controls. The calibrated stack has the lowest observed follow-up Brier, 0.13913 versus 0.14038 for calibrated BN, but its conditional interval includes zero and statistically significant superiority is not established. OOF-to-full-fit shift and post-selection uncertainty qualify this exploratory result; it is neither external validation nor a novel algorithm. The comparison remains an academic prediction exercise with preserved negative findings.

XI. Authorship and Accountability

The five author names were supplied by the team representative and the institution matches the Phase 1 report. A proposed, approximately balanced responsibility allocation is maintained separately in the repository for member confirmation. It is not evidence of completed work, and no equal-contribution or individual completed-work statement is asserted here. Each author should confirm actual contributions and review the complete submission before final hand-in.

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